2 SET THEORY

Learning outcomes

By the end of this topic you should be able to:

2.1 Represent a set using **extensive notation** and determine if two sets are **equal** 2-1
2.2 Represent a set using **set builder** (intensive) notation 2-3
2.3 Understand the concepts of **empty set** and **universal set** 2-4
2.4 Determine if one set is a **subset** of another 2-5
2.5 Understand and apply common **set operations** 2-6
2.6 Use **Venn diagrams** to solve problems involving three sets 2-9
2.7 Use **fundamental set identities** to show the equality of two sets 2-13
2.8 Determine the **cardinality** of a set 2-15
2.9 Determine the **power set** of a set 2-17
2.10 Determine the **Cartesian product** of two sets 2-18
2.11 Determine **partitions** of a set 2-18

2.1 Represent a set using **extensive notation** and determine if two sets are **equal**

SET

A **set** is a well-defined[a] collection of objects. The objects in a set are called the **members** of the set.

[a] This means it is unambiguous as to whether an object is a member of a set.

EXTENSIVE NOTATION

One way to write down a set is to write down its members, separated by commas and enclosed between curly braces { }. For example $\{1, z, \text{red}\}$ is the set containing the three objects 1, z and 'red'.

Explicitly listing the members of a set between curly braces in this way is called **extensive notation** (aka *roster* notation).

EQUALITY OF SETS

Two sets are **equal** if and only if they have exactly the same members.

$\{a, b\} = \{b, a\}$

The order in which the members of a set are listed between the curly braces { } does not matter. For example, $\{1, z, \text{red}\} = \{z, \text{red}, 1\}$

$\{x\} \neq x$

The set containing the object x is not the same as the object x.

$\{a, a\} = \{a\}$

An object cannot occur more than once in a set.

$\in, \notin$

If a is a member of set A, then we write this as $a \in A$.
If a is not a member of set A, then we write this as $a \notin A$.
For example, $5 \in \{1, 2, 3, 4, 5\}$ and $8 \notin \{2, 3, 4\}$.

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Worked Examples

EXAMPLE 2.1

Let $A = \{x\}$. Are the following objects members of A?

2.1.1 x	**Solution** A is a set with 1 member, x. So $x \in A$.
2.1.2 $\{x\}$	**Solution** $\{x\}$ is the set containing x; this is not a member of A. So $\{x\} \notin A$.

EXAMPLE 2.2

Consider the set $A = \{1, 2, 5\}$. Which of the following sets are equal to A?

2.2.1 $X = \{2, 5, 1\}$	**Solution** $\{2, 5, 1\} = \{1, 2, 5\}$, as the order of the members does not matter. So $X = A$.
2.2.2 $Y = \{1, 5, 1, 2, 2, 1\}$	**Solution** $\{1, 5, 1, 2, 2, 1\} = \{1, 2, 5\}$, as multiple instances of the same object cannot exist in a set. So $Y = A$.
2.2.3 $Z = \{\{1, 2, 5\}\}$	**Solution** Z is the set with A as a member. So $Z \neq A$.

Classroom Exercises

EXERCISE 2.1

Which of the following are valid sets?

2.1.1 The set of days of the week.	**Solution** Yes, each member is well defined.
2.1.2 The set of all tall men.	**Solution** No: 'tall' is not well-defined.
2.1.3 The set of all prime numbers.	**Solution** Yes. Although we do not know every member, each member is well defined.

EXERCISE 2.2

Write the following sets using extensive notation.

2.2.1 The set of vowels in the English alphabet.	**Solution** $\{a, e, i, o, u\}$
2.2.2 The set of prime numbers less than 10.	**Solution** $\{2, 3, 5, 7\}$
2.2.3 The set of digits in the number 'one million and one'.	**Solution** $\{1, 0\}$

EXERCISE 2.3

Let $X = \{1, \{2\}, \{1, \{1, 2\}\}\}$. Are the following objects members of X?

2.3.1 1	**Solution** Yes, $1 \in X$.
2.3.2 2	**Solution** No, $2 \notin X$.
2.3.3 $\{1, 2\}$	**Solution** No, $\{1, 2\} \notin X$. Note that $\{1, 2\}$ is a member of $\{1, \{1, 2\}\}$, which is a member of X; but $\{1, 2\}$ is not itself a member of X.

2.2 Represent a set using **set builder** (intensive) notation

SET-BUILDER NOTATION

Instead of listing the members of a set explicitly using extensive notation, we can instead specify those properties that characterize the members in the set. This is called **Intensive notation** or **set builder notation**.

For example, the set X of positive integers which are less than or equal to 1000 can be written using intensive notation as $X = \{x : x \in \mathbb{Z}, 0 < x \leq 1000\}$.

This is read as 'X is the set of x such that x is an integer and greater than 0 and less than or equal to 1000'. The letter x denotes a typical member of the set; the colon is read as 'such that' or 'where', and the comma as 'and'.

Worked Examples

EXAMPLE 2.3

Write the following sets using set-builder notation.

2.3.1 $\{-1, -2, -3, -4, \ldots\}$	**Solution** $\{-n : n \in \mathbb{N}\}$
2.3.2 The set of odd integers.	**Solution** $\{2n + 1 : n \in \mathbb{Z}\}$

Classroom Exercises

EXERCISE 2.4

Write the following sets using set-builder notation.

2.4.1

$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Solution

$\{n : n \in \mathbb{N}, n \leq 10\}$

2.4.2

The set of square numbers.

Solution

$\{n^2 : n \in \mathbb{Z}\}$

2.3 Understand the concepts of **empty set** and **universal set**

EMPTY SET $\emptyset$

The set with no members is called the **null set** or **empty set**. It is denoted by $\emptyset$. Thus, $\emptyset = \{\}$. There is only one empty set, i.e. if A and B are both empty sets, then $A = B$.

UNIVERSAL SET $\mathcal{U}$

The members of all sets under consideration in a particular problem belong to some fixed (sometimes large) set called the **universal set**. The universal set is denoted by $\mathcal{U}$.

The universal set varies from question to question. Often the universal set will be $\mathbb{N}$ or $\mathbb{Z}$.

Worked Examples

EXAMPLE 2.4

Give an example of the empty set expressed using set-builder notation.

Solution Any valid example, e.g.:

- $\{x \in \mathbb{N} : x < 0\}$
- $\{x : x$ is a county in Northern Ireland beginning with the letter 'z'$\}$
- $\{x : x$ is a number that is both even and odd.$\}$

Classroom Exercises

EXERCISE 2.5

Which (if any) of the following sets are the same? $\emptyset$, $\{\emptyset\}$, $\{\}$ and $\{0\}$.

Solution

- $\emptyset$ is the set with zero members.
- $\{\emptyset\}$ is a set with one member, namely the empty set.
- $\{0\}$ is a set with one member, namely the number 0.
- $\{\}$ is the set with zero members.

Therefore $\emptyset$ and $\{\}$ are the only sets which are the same.

2.4 Determine if one set is a **subset** of another

SUBSET, $A \subseteq B$

If every member of a set A is also a member of a set B, then A is called a **subset** of B. The notation for this is $A \subseteq B$.

Also, B is called a superset of A. The notation for this is $B \supseteq A$.

If A has at least one member that is not a member of B, then A is not a subset of B, and we write $A \nsubseteq B$, or $B \nsupseteq A$.

PROPER SUBSET

$\subseteq$ and $\supseteq$ are used when there is the possibility that A and B are the same set. If B has members which are not members of A, then A is called a proper subset of B, and we use $\subset$ and $\supset$.

$A \subseteq A$

Every set is a subset of itself.

$\emptyset \subseteq A$

The empty set is a subset of every set.

If $A \subseteq B$ and $B \subseteq A$, then $A = B$.

If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Worked Examples

EXAMPLE 2.5

Suppose $A = \{a, b, c, d, e\}$, $B = \{b, c, d\}$, $C = \{c, d, e, f\}$.

2.5.1 Is $B \subseteq A$?	**Solution** $B \subseteq A$, because every member of B is also a member of A.
2.5.2 Is $C \subseteq A$?	**Solution** $C \nsubseteq A$, because f is a member of C but is not a member of A.

Classroom Exercises

EXERCISE 2.6

Let $A = \{2, 4, 6\}$, $B = \{2, 6\}$, $C = \{4, 6\}$ and $D = \{4, 6, 8\}$. Are each of the following true or false?

2.6.1 $B \subseteq A$	**Solution** True
2.6.2 $C \subseteq A$	**Solution** True
2.6.3 $D \subseteq A$	**Solution** False, as $8 \notin A$
2.6.4 $B \subseteq C$	**Solution** False, as $2 \notin C$
2.6.5 $D \subseteq B$	**Solution** False, as $6, 8 \notin B$
2.6.6 $C \subseteq D$	**Solution** True

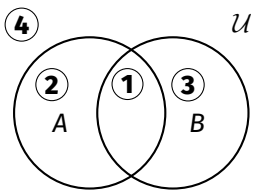
2.5 Understand and apply common **set operations**

VENN DIAGRAM

To visualise operations on sets, we will use **Venn diagrams**. In a Venn diagram, each set is represented by a circle[a]; the members of that set then lie inside that circle. The circles themselves are drawn inside a rectangle, which represents the universal set $\mathcal{U}$.

[a] Circles can be used for two and three set Venn diagrams; for more than 3 sets we need different shapes.

2-set Venn diagram



Suppose we have two arbitrary sets A and B. By representing A and B as two overlapping circles inside a rectangle representing $\mathcal{U}$ (as shown), we can see that any object $x \in \mathcal{U}$ must be placed in one of the four regions ① - ④. These four regions represent all the possibilities regarding x's membership of A and B:

① $x \in A$ and $x \in B$

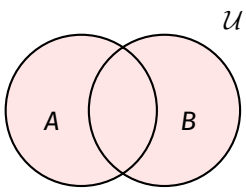
② $x \in A$ and $x \notin B$

③ $x \notin A$ and $x \in B$

④ $x \notin A$ and $x \notin B$

Below we define common **operations** on sets. Most of these operations are binary, i.e. they operate on a pair of sets A and B, and return a new set. For each operation described, we display the result of the operation as a shaded region on a Venn diagram.

Union $A \cup B$

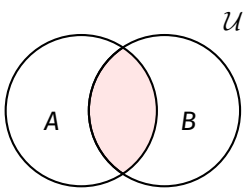


The **union** of two sets, A and B, denoted by $A \cup B$, is the set of all elements which belong to A or B:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

The union of n sets $A_1, A_2, \ldots, A_n$ is denoted $\bigcup_{i=1}^{i=n} A_i$

Intersection $A \cap B$



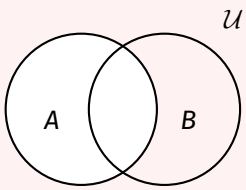
The **intersection** of two sets, A and B, denoted by $A \cap B$, is the set of all elements which belong to both A and B.

The intersection of n sets $A_1, A_2, \ldots, A_n$ is denoted as $\bigcap_{i=1}^{i=n} A_i$

Disjoint sets

If $A \cap B = \emptyset$, then A and B do not have any elements in common. In this case, A and B are said to be **disjoint**.

Complement A^c

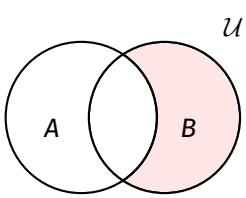


The **complement** of the set A, denoted by A^c is the set of elements which belong to the universal set $\mathcal{U}$ but which do not belong to A:

$$A^c = \{x : x \in \mathcal{U}, x \notin A\}$$

Common alternative notations for complement are $A^{\sim}$, A' and $\overline{A}$.

Relative complement $B - A$

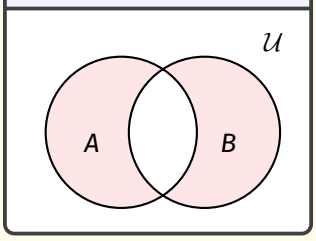


The complement of set A **relative to set** B, denoted $B - A$, is the set of members of B which are not members of the set A.

$$B - A = \{x : x \in B \text{ and } x \notin A\}$$

$B \setminus A$ is an alternative notation which is commonly used for relative complement.

SYMMETRIC DIFFERENCE
$A \triangle B$



The **symmetric difference** of sets A and B, denoted by $A \triangle B$, consists of those elements which belong to A or B, but not to both. That is,

$$A \triangle B = \{x : (x \in A \text{ and } x \notin B) \text{ or } (x \notin A \text{ and } x \in B)\}$$

Equivalently

$$A \triangle B = (A \cup B) - (A \cap B)$$

$A \oplus B$ is alternative notation which some use for symmetric difference.

Worked Examples

EXAMPLE 2.6

Let $A = \{1, 3, 5, 7, 9\}$, $B = \{3, 6, 9, 12\}$, and $C = \{x \in \mathbb{N} : 5 \leq x \leq 9\}$. List the members of the following sets:

2.6.1 $(A \cap B) \cap C$	**Solution** $(A \cap B) \cap C = \{3, 9\} \cap \{5, 6, 7, 8, 9\} = \{9\}$
2.6.2 $(A \cap B) \cup (B \cap C)$	**Solution** $(A \cap B) \cup (B \cap C) = \{3, 9\} \cup \{6, 9\} = \{3, 6, 9\}$
2.6.3 $C - B$	**Solution** $C - B = \{5, 7, 8\}$
2.6.4 $A \triangle B$	**Solution** $A \triangle B = \{1, 5, 6, 7, 12\}$

Classroom Exercises

EXERCISE 2.7

Let $A = \{1, 2, 3, 4, 5\}$, $B = \{x \in \mathbb{N} : 3 \leq x \leq 8\}$ and $C = \{2, 4, 6, 8\}$. List the members of the following sets:

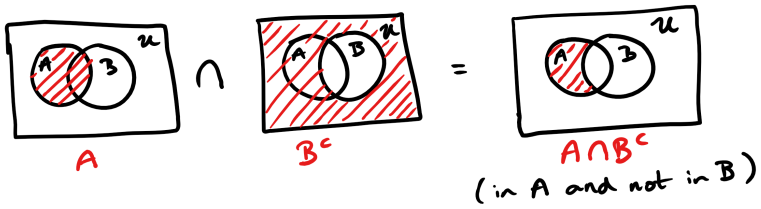
2.7.1 $A \cup B$	**Solution** $\{1, 2, 3, 4, 5, 6, 7, 8\}$
2.7.2 $(A \cap B) \cap C$	**Solution** $\{3, 4, 5\} \cap \{2, 4, 6, 8\} = \{4\}$
2.7.3 $(A \cap B) \cup (B \cap C)$	**Solution** $\{3, 4, 5\} \cup \{3, 4, 6, 8\} = \{3, 4, 5, 6, 8\}$
2.7.4 $A \triangle B$	**Solution** $(A \cup B) \setminus (A \cap B) = \{1, 2, 3, 4, 5, 6, 7, 8\} \setminus \{3, 4, 5\} = \{1, 2, 6, 7, 8\}$

EXERCISE 2.8

On a Venn diagram for two sets A and B, shade the regions representing:

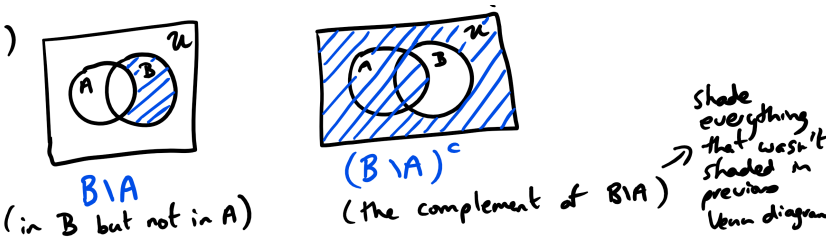
2.8.1 $A \cap B^c$

Solution



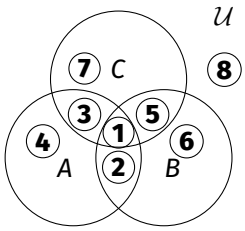
2.8.2 $(B \setminus A)^c$

Solution



2.6 Use **Venn diagrams** to solve problems involving three sets

3-SET VENN DIAGRAM



Suppose we want to model three sets A, B and C using a Venn diagram. Consider an arbitrary object $x \in \mathcal{U}$. There are eight different possibilities for x's membership to A, B and C. These 8 possibilities are shown in the truth table below, and labelled in the Venn diagram shown.

$x \in A$	$x \in B$	$x \in C$	region
T	T	T	1
T	T	F	2
T	F	T	3
T	F	F	4
F	T	T	5
F	T	F	6
F	F	T	7
F	F	F	8

Worked Examples

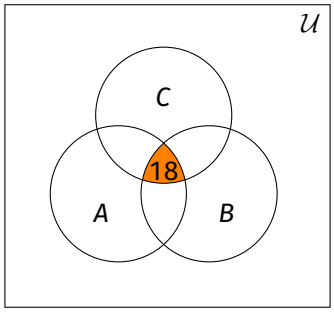
EXAMPLE 2.7

In the University sports centre there are 530 members in the athletics club, 210 members in the basketball club and 180 members in the cricket club. Of them, 100 are members of both the athletics and basketball clubs, 68 are in both athletics and cricket clubs, 53 are both basketball and cricket clubs and 18 are in all three sports clubs.

How many members are there in the athletics club only? What is the total number of members across the three sport clubs?

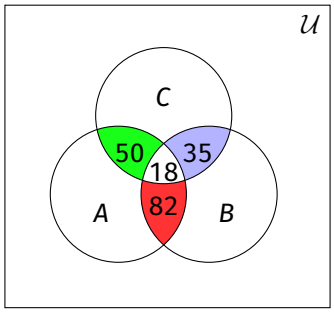
Solution We refer to the athletics club as set A, the basketball club as set B and the cricket club as set C, and solve using a 3-set Venn diagram.

Step 1 Start by considering the orange coloured region at the centre of the Venn diagram. This region is the overlap of the three sets, $A \cap B \cap C$. It represents those people who are in all three sports clubs. There are 18 people in all three sports club, so write 18 here.



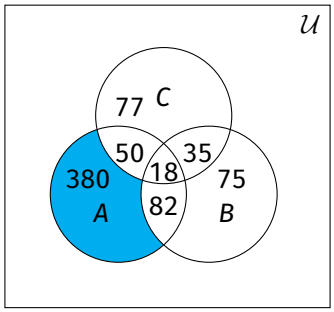
Step 2 100 people are members of both the athletics club and the basketball club. These 100 are spread across the central region (containing the 18 people who are members of all clubs) and the red shaded region (members of athletics and basketball, but not cricket). As there are 18 people in the central region, there must be 100 - 18 = 82 people in the red shaded region.

Similar logic tells us that there must be 68-18 = 50 people in the green shaded region, and 53-18=35 people in the blue shaded region.



Step 3 There are 530 in the athletics club. They are all contained inside circle A, which is made up of four regions. We have already determined the number of people in three of these regions. The remaining cyan coloured region contains people who are members of the athletics club only; the number here must be $530 - (82 + 18 + 50) = 380$.

Similar logic tells us the number of people who play basketball only is $210 - (18 + 82 + 35) = 75$, and cricket only is $180 - (18 + 50 + 35) = 77$.



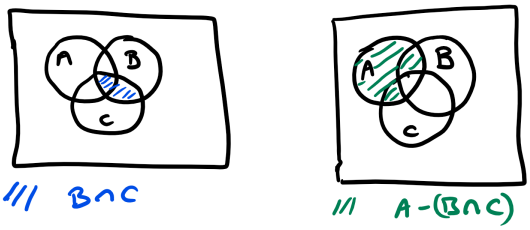
So: 380 are in the athletics club only. The total number of members across the 3 sports clubs is $380 + 75 + 77 + 35 + 50 + 82 + 18 = 717$

Classroom Exercises

Exercise 2.9

For any 3 non-empty sets A, B and C draw Venn diagrams to determine which (if any) of the following is equivalent to $A - (B \cap C)$:

Venn diagram for $A-(B\cap C)$:



2.9.1

$(A - B) \cap (A - C)$

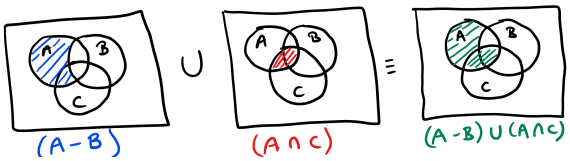
Solution

Different regions are shaded compared to $A-(B\cap C)$, so $(A - B) \cap (A \cap C) \not\equiv A - (B \cap C)$

2.9.2

$(A - B) \cup (A \cap C)$

Solution



Different regions are shaded compared to $A-(B\cap C)$, so $(A - B) \cup (A - C) \not\equiv A - (B \cap C)$

2.9.3

$(A - B) \cup (A - C)$

Solution

The same regions are shaded as for $A - (B \cap C)$, so $(A - B) \cup (A - C) \equiv A - (B \cap C)$

2.9.4

$(A - B) - (A - C)$

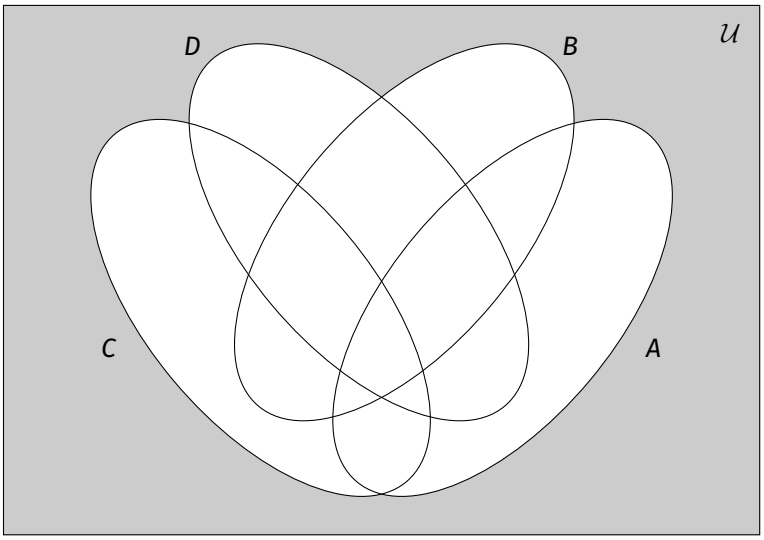
Solution

Different regions are shaded compared to $A-(B\cap C)$, so $(A - B) - (A - C) \not\equiv A - (B \cap C)$

EXERCISE 2.10

Difficult: How many distinct regions does a 4-set Venn diagram need? Try to draw it.

Solution A 4-set Venn diagram needs $2^4 = 16$ distinct regions. It is impossible to draw using circles to represent the sets. Below is one version using ellipses:



See http://www.newscientist.com/gallery/venn/ for examples of Venn diagrams involving more than 4 sets.

2.7 Use **fundamental set identities** to show the equality of two sets

There are several **fundamental set identities** which may be used to rewrite expressions involving set operations in equivalent forms.

COMMUTATIVITY

$A \cup B \equiv B \cup A$
$A \cap B \equiv B \cap A$

ASSOCIATIVITY

$A \cup (B \cup C) \equiv (A \cup B) \cup C$
$A \cap (B \cap C) \equiv (A \cap B) \cap C$

DISTRIBUTIVITY

$A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$
$A \cup (B \cap C) \equiv (A \cup B) \cap (A \cup C)$

DE MORGAN'S LAWS

$(A \cup B)^c \equiv A^c \cap B^c$
$(A \cap B)^c \equiv A^c \cup B^c$

IDEMPOTENCY	$A \cup A \equiv A$ $A \cap A \equiv A$
INVOLUTION LAW	$(A^c)^c \equiv A$
COMPLEMENT LAWS	$A \cup A^c \equiv \mathcal{U}$ $A \cap A^c \equiv \emptyset$
IDENTITY LAWS	$A \cup \mathcal{U} \equiv \mathcal{U}$ $A \cap \mathcal{U} \equiv A$ $A \cup \emptyset \equiv A$ $A \cap \emptyset \equiv \emptyset$
COMPLEMENTS OF $\mathcal{U}$ AND $\emptyset$	$\mathcal{U}^c \equiv \emptyset$ $\emptyset^c \equiv \mathcal{U}$
ABSORPTION LAWS	$A \cup (A \cap B) \equiv A$ $A \cap (A \cup B) \equiv A$
DUALITY PRINCIPLE	Any valid identity has an equally valid 'dual' found by replacing: $\cap$ with $\cup$, $\cup$ with $\cap$, $\mathcal{U}$ with $\emptyset$ and $\emptyset$ with $\mathcal{U}$

Worked Examples

EXAMPLE 2.8

Use the fundamental set identities to show that:

$$(A \cap B)^c \cap C \equiv (A^c \cap C) \cup (B^c \cap C)$$

Solution

$$\begin{aligned}
(A \cap B)^c \cap C &\equiv (A^c \cup B^c) \cap C && \text{by de Morgan} \\
&\equiv C \cap (A^c \cup B^c) && \text{by commutativity} \\
&\equiv (C \cap A^c) \cup (C \cap B^c) && \text{by distributivity} \\
&\equiv (A^c \cap C) \cup (B^c \cap C) && \text{by commutativity}
\end{aligned}$$

Classroom Exercises

EXERCISE 2.11

Use the fundamental set identities to prove $(A \cup B) \cap (A \cup B^c) \equiv A$

Solution

$$
\begin{aligned}
&(A \cup B) \cap (A \cup B^c) \\
&\equiv A \cup (B \cap B^c) && \text{by distributivity} \\
&\equiv A \cup \phi && \text{by complement law} \\
&\equiv A && \text{by identity law}
\end{aligned}
$$

EXERCISE 2.12

What is the dual of $(A \cup \mathcal{U})^c \cap A \equiv \emptyset$ **?**

Solution

$$(A \cup \mathcal{U})^c \cap A \equiv \phi$$

DUAL: $(A \cap \phi)^c \cup A \equiv \mathcal{U}$

2.8 Determine the **cardinality** of a set

CARDINALITY

The **cardinality** of a set A, denoted by $|A|$, is the number of elements contained in A. For example, if $A = \{1, 4, 5\}$ then $|A| = 3$.

PRINCIPLE OF INCLUSION AND EXCLUSION

The **principle of inclusion and exclusion** is a method for determining the cardinality of a union of finite sets. It states:

- For two finite sets A and B:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

- For three finite sets A, B and C:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Worked Examples

EXAMPLE 2.9

Determine the cardinality of each of the following sets:

2.9.1 $\{\}$	**Solution** $\lvert\{\}\rvert = 0$
2.9.2 $\{1, \{2, 3, 3\}\}$	**Solution** $\lvert\{1, \{2, 3, 3\}\}\rvert = 2$
2.9.3 $\{5, 5, 5, 5, 5\}$	**Solution** $\lvert\{5, 5, 5, 5, 5\}\rvert = \lvert\{5\}\rvert = 1$
2.9.4 $\{\{0\}, \emptyset, \{\}, 0, \{\emptyset\}\}$	**Solution** $\emptyset, \{\}, 0, \{\emptyset\}\}\rvert = \lvert\{\{0\}, \emptyset, 0, \{\emptyset\}\}\rvert = 4$

EXAMPLE 2.10

Calculate $|A \cup B|$, where $A = \{a, b, c, d, e\}$ and $B = \{d, e, f, g, h, i\}$

Solution

- $|A| = 5$, $|B| = 6$ and $|A \cap B| = |\{d, e\}| = 2$.
- Therefore $|A \cup B| = |A| + |B| - |A \cap B| = 5 + 6 - 2 = 9$.

Classroom Exercises

EXERCISE 2.13

Determine the cardinality of each of the following sets.

2.13.1 $\{\emptyset\}$	**Solution** $\lvert\{\emptyset\}\rvert = 1$
2.13.2 $\{x \in \mathbb{N} : x < 100,$ x contains the digit $1\}$	**Solution** $\lvert\{x \in \mathbb{N} : x < 100, x \text{ contains the digit } 1\}\rvert$ $= \lvert\{1, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 21, 31, 41, 51, 61, 71, 81, 91\}\rvert = 19$
2.13.3 $\{3, 3, \{3\}\}$	**Solution** $\lvert\{3, 3, \{3\}\}\rvert = \lvert\{3, \{3\}\}\rvert = 2$
2.13.4 $\{\mathbb{N}\}$	**Solution** $\lvert\{\mathbb{N}\}\rvert = 1$

EXERCISE 2.14

Suppose $|A| = 15$, $|A \cap B| = 12$ and $|A \cup B| = 22$. What is the value of $|B|$?

Solution Rearranging $|A \cup B| = |A| + |B| - |A \cap B|$ gives $|B| = |A \cup B| + |A \cap B| - |A|$. So $|B| = 22 + 12 - 15 = 19$.

EXERCISE 2.15

In a class of 30 students, 9 have an A-level in maths, 14 have an A-level in physics, and 5 have an A-level in both maths and physics. How many students in the class do not have an A-level in either maths or physics?

Solution Let A be the set of students with an A-level in maths, and B be the set of students with an A-level in physics. The number of students with an A-level in maths or physics is $|A \cup B| = |A| + |B| - |A \cap B| = 9 + 14 - 5 = 18$. So the number of students without an A-level in either is $30 - 18 = 12$.

2.9 Determine the **power set** of a set

POWER SET

The set of all the subsets of A is called the power set of A. It is denoted $\mathcal{P}(A)$.

The cardinality of the power set of A is $\mathcal{P}(A) = 2^{|A|}$.

Worked Examples

EXAMPLE 2.11

What is $\mathcal{P}(X)$ and $|\mathcal{P}(X)|$ when $X = \{1, 2\}$?

Solution Set X has:

- 1 subset with 2 elements, namely $\{1, 2\}$.
- 2 subsets with 1 element, namely $\{1\}$ and $\{2\}$.
- 1 subset with 0 elements, namely $\emptyset$.

Therefore $\mathcal{P}(X) = \{\{1, 2\}, \{1\}, \{2\}, \emptyset\}$ and $|\mathcal{P}(X)| = 4$.

Classroom Exercises

EXERCISE 2.16

What is the cardinality of the power set of A when:

2.16.1 $A = \emptyset$	**Solution** $\lvert A\rvert = 0$ so $\lvert\mathcal{P}(A)\rvert = 2^0 = 1$
2.16.2 $A = \{\emptyset\}$	**Solution** $\lvert A\rvert = 1$ so $\lvert\mathcal{P}(A)\rvert = 2^1 = 2$
2.16.3 $A = \{x \in \mathbb{N} : 11 \leq x \leq 20\}$	**Solution** $\lvert A\rvert = 10$ so $\lvert\mathcal{P}(A)\rvert = 2^{10} = 1024$
2.16.4 $A = \{a, e, i, o, u, i, e\}$	**Solution** $\lvert A\rvert = 5$ so $\lvert\mathcal{P}(A)\rvert = 2^5 = 32$

2.10 Determine the **Cartesian product** of two sets

ORDERED PAIR

An **ordered pair** consists of two objects listed inside parentheses (). The order of the object matters, e.g. (1,2) is different from (2,1).

CARTESIAN PRODUCT

The **Cartesian product** of two sets A and B, written $A \times B$, is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$. So

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

The Cartesian product of n sets $A_1, A_2, \dots, A_n$ is written as $\bigtimes_{i=1}^{i=n} A_i$.

It is the set of all ordered n-tuples $(a_1, a_2, \dots, a_n)$ where $a_1 \in A_1, a_2 \in A_2, \dots, a_n \in A_n$.

Worked Examples

EXAMPLE 2.12

What is $A \times B$ if $A = \{a, b\}$ and $B = \{1, 2, 3\}$?

Solution

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

Classroom Exercises

EXERCISE 2.17

If $A = \{2, 7\}$ and $B = \{5, 6\}$, find:

2.17.1 $A \times B$	**Solution** $\{(2, 5), (2, 6), (7, 5), (7, 6)\}$
2.17.2 $B \times A$	**Solution** $\{(5, 2), (5, 7), (6, 2), (6, 7)\}$
2.17.3 $A \times A$	**Solution** $\{(2, 2), (2, 7), (7, 2), (7, 7)\}$
2.17.4 $B \times B \times B$	**Solution** $\{(5, 5, 5), (5, 5, 6), (5, 6, 5), (5, 6, 6), (6, 5, 5), (6, 5, 6), (6, 6, 5), (6, 6, 6)\}$

2.11 Determine **partitions** of a set

PARTITION

Let S be a non-empty set. A **partition** of S is a subdivision of S into non-overlapping, non-empty subsets.

More precisely, a partition of S is a collection $\{A_i\}$ of non-empty subsets of S such that:

1. Each $a \in S$ belongs to one of the A_i
2. The sets of A_i are mutually disjoint; that is, if $A_i \neq A_j$ then $A_i \cap A_j = \emptyset$.

The subsets in a partition are called cells.

Worked Examples

EXAMPLE 2.13

Which of the following is a partition of $X = \{1, 2, 3, 4, 5, 6, 7, 8\}$?

2.13.1

$\{\{1, 3, 5\}, \{2, 4\}, \{7, 8\}\}$

Solution

This is not a partition of X, as $6 \in X$, but 6 does not belong to any of the cells.

2.13.2

$\{\{1, 3, 5\}, \{2, 3, 4\}, \{6, 7, 8\}\}$

Solution

This is not a partition of X, as $\{1, 3, 5\} \cap \{2, 3, 4\} = \{3\} \neq \emptyset$, i.e the cells are not disjoint.

2.13.3

$\{\{1, 3, 5\}, \{2, 4, 6\}, \{7, 8\}\}$

Solution

This is a partition of X.

Classroom Exercises

EXERCISE 2.18

Find all the partitions of $A = \{1, 2, 3\}$.

Solution

- $\{\{1\}, \{2\}, \{3\}\}$
- $\{\{1, 2\}, \{3\}\}$
- $\{\{1, 3\}, \{2\}\}$
- $\{\{2, 3\}, \{1\}\}$
- $\{\{1, 2, 3\}\}$

Further resources

Text books

- Section 2.2–2.4 of W. D. Wallis (2012), *A Beginner's Guide to Discrete Mathematics*, 2nd ed, Birkhäuser.
- Chapter 1 of D. Mackinson (2020), *Sets, Logic and Maths for Computing*, 3rd ed, Springer.
- Section 4.1–4.4 of J. P. Fortney (2020), *Discrete Mathematics for Computer Science*, 1st ed, CRC Press.
- Section 2.2 of G. O'Regan (2016), *Guide to Discrete Mathematics*, Springer.

Youtube videos

Link	Learning outcome										
	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	2.10	2.11
GVSUmath	x										
MyWhyU	x	x	x								
GVSUmath		x									
MyWhyU				x							
HCCMathHelp				x							
PatrickJMT					$\cup, \cap$						
lbowen11235					$\cup, \cap, A^C$						
Khan Academy					$\cup, \cap$						
MyWhyU					$B \setminus A$						
Khan Academy					$B \setminus A$						
MyWhyU					$A \triangle B$						
patrickJMT (1/2)						x					
patrickJMT (2/2)						x					
FREGE							x				
stoleemath							x				
Mike Peller									x		
Stats-Lab Dublin									x		
MyWhyU										x	